

Talk-to-Me Drone

Weekly Report - 2026-07-23

FocusEvaluation framework, control-loop validation, intent redesign, multi-platform robustness, and cross-language planning.

1. Core Change To Discuss This Week

The story can be framed by the following question:

When a drone is already operating, can a spoken input pass through speech processing, intent recognition, drone-control policy, and drone response as a reproducible, measurable, and explainable control loop?

This framing separates the problem into four parts:

- Input side:** whether speech remains reliable under rotor noise, distance changes, and volume changes.
- Recognition side:** whether different intent categories can be separated robustly.
- Control side:** whether recognized intent can pass through drone-control policy and trigger the correct drone response.
- Experimental side:** whether the same workflow can be repeated across platforms and conditions.

2. Revised Evaluation Framework

Layer	Question To Answer	Evidence To Collect
Physical acoustic operating range	Under what real physical conditions can the system hear speech?	SNR, speech volume, distance, microphone placement, clipping rate
Improved intent recognition	Can the redesigned intent categories be recognized reliably?	Per-intent accuracy/recall, confusion matrix, confidence distribution
Complete control loop	After takeoff, can the drone receive a command and produce the corresponding response?	Command issued, policy decision, drone acknowledgement, observed response, video/log alignment, latency
Multi-platform robustness	How do drone model, propeller type, and hardware placement affect the system?	Drone model, rotor spectrum, SNR, recognition metrics, control-loop metrics

3. Controlled Playback Before User Study

Before formal user studies, we should use controlled playback to validate the protocol. The goal is to use the same speech content and vary one physical variable at a time.

Initial setup:

- One fixed drone platform.
- One fixed microphone placement rule.
- One unified playback source.
- Initial distances such as 1 m and 2 m.
- Conversational and projected volume levels.
- Rotor off and rotor on conditions.
- Metadata, logs, video, SNR or SPL, and failure notes saved for every run.

The purpose of this pilot is not to obtain polished final results. The purpose is to check whether the protocol is clear, repeatable, and easy for the advisor and Stephen to inspect before we scale up data collection.

4. Intent Categories Need To Be Redesigned

The current **emergency / movement / unknown** taxonomy is not suitable for a complete control-loop evaluation. In particular, **movement** only says that the user wants the drone to move; it does not specify direction, distance, speed, or action type.

A first-round control-loop taxonomy could be:

Intent Group	Example Commands	Drone-Side Meaning
Emergency / stop	stop, emergency, abort	Immediately enter hold, land, or emergency handling
Takeoff / landing	take off, land	Change the flight state
Hover / hold	hold, stay, pause	Maintain position or stop the current action
Directional movement	forward, back, left, right, up, down	Small controlled directional motion
Yaw / orientation	turn left, turn right	Change heading or orientation
Unknown / unsupported	unrelated speech, unsupported phrase	No action or reject

5. Multi-Platform Evaluation

A practical design is two-layered:

- 1
- Primary platform (Tello) deep evaluation:** run the full physical matrix and control-loop evaluation on the current main drone.
- 2
- Secondary platform transfer check:** test one or two additional drones, propellers, or mounting configurations with the same prompt set, distance/volume settings, and microphone-placement rule. This does not need to repeat the full matrix at first.

Metrics:

- Rotor-noise spectrum and SNR difference.
- Recognition accuracy, emergency recall, and unknown false admission.
- Control-loop response success.
- Latency, acknowledgement, and timeout.
- Platform-specific failure cases.
- Microphone-placement sensitivity.

6. Cross-Language

Cross-language should remain in the plan, but it should serve the redesigned intent taxonomy and control-loop story. It should not be framed as simply asking whether the model can recognize another language.

Candidate data sources:

Dataset	Use	Initial Judgment
MLCommons Multilingual Spoken Words Corpus	Keyword-level multilingual command or KWS-style testing	Good fit for direction words, stop/help, and similar command keywords
Mozilla Common Voice	Transcript filtering, phrase-level speech, unknown/general speech	Good fit for phrase construction and unknown examples
FLEURS	102-language general speech and language robustness reference	Useful as unknown/general speech reference
MInDS-14	Multilingual spoken intent benchmark	Better suited for related work or sanity checking than as the main drone dataset

Key design issue: Each language should map into the same drone-side taxonomy, and unsupported phrases should remain unknown rather than being forced into actionable classes.

Source note

This PDF is a faithful export of the finalized English Notion report dated 2026-07-23. [Open the source page.](#)